Part 3

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Iteration 0

* features (X) : $x = ((1,3),(4,10))$

Let $X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$

* targets : $y = (5,6) \Rightarrow y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$

* weights (m) : $m = [-1,2] \Rightarrow m = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$

* bias (b) : $b = [1,1] \Rightarrow b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

* Number of samples : $N = 2$

* learning rate : $\alpha = 0.01$

$$y = m_1 x_1 + m_2 x_2 + b$$

1. Compute predicted values $\hat{y}$

$$\hat{y} = mX + b$$

* calcule mX

$$mX = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1\times(-1)+(3\times 2) \\ 4(-1)+(10\times 2) \end{bmatrix} = \begin{bmatrix} -1+6 \\ -4+20 \end{bmatrix} = \begin{bmatrix} 5 \\ 16 \end{bmatrix}$$

$$\Rightarrow \boxed{mX = \begin{bmatrix} 5 \\ 16 \end{bmatrix}}$$

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$$\hat{y} = mX + b \Rightarrow \hat{y} = \begin{bmatrix} 5 \\ 16 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\boxed{\hat{y} = \begin{bmatrix} 6 \\ 17 \end{bmatrix}}$$

* calculate error

$$\hat{y} - y = \begin{bmatrix} 6 \\ 17 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$$\boxed{\hat{y} - y = \begin{bmatrix} 1 \\ 11 \end{bmatrix}}$$

* MSE

$$J = \frac{1}{2}(1^2 + 11^2) = \frac{122}{2} = 61 \Rightarrow \boxed{J = 61}$$

2. Derive the gradient using the chain rule

$$J = \frac{1}{N}\sum(\hat{y} - y)^2$$

Chain rule for m $\left(\frac{\partial J}{\partial m}\right)$

$$\frac{\partial J}{\partial m} = \frac{\partial J}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial m}$$

$$\frac{\partial J}{\partial \hat{y}} = \left(\frac{1}{N}(\hat{y} - y)^2\right)'$$

$$\frac{\partial J}{\partial \hat{y}} = 2 \times \frac{1}{N} \times 1(\hat{y} - y)$$

$$\Rightarrow \boxed{\frac{\partial J}{\partial \hat{y}} = \frac{2}{N}(\hat{y} - y)}$$

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$$\hat{y} - y = \begin{bmatrix} 6 \\ 17 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

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Chain rule for m $\left(\frac{\partial J}{\partial m}\right)$

(*) $$\frac{\partial J}{\partial m} = \frac{\partial J}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial m}$$

(a) outer derivative

$$\frac{\partial J}{\partial \hat{y}} = \left(\frac{1}{N}(\hat{y} - y)^2\right)'$$

$$\frac{\partial J}{\partial \hat{y}} = 2 \times \frac{1}{N} \times 1(\hat{y} - y)$$

$$\Rightarrow \boxed{\frac{\partial J}{\partial \hat{y}} = \frac{2}{N}(\hat{y} - y)}$$

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(b) inner derivative

$$\frac{\partial \hat{y}}{\partial b} = 1$$

(c) $$\frac{\partial J}{\partial b} = 1 \times \frac{2}{N}(\hat{y} - y)$$

$$\boxed{\frac{\partial J}{\partial b} = \frac{2}{N}(\hat{y} - y)}$$

3. Calcule numerical gradients

* gradient for m : $X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$

$$X^T = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix}$$

$$\frac{\partial J}{\partial m} = \frac{2}{2}\begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix}\begin{bmatrix} 1 \\ 11 \end{bmatrix} = 1 \times \begin{bmatrix} 1\times1 + (4\times11) \\ (3\times1) + (10\times11) \end{bmatrix} = \begin{bmatrix} 1+44 \\ 3+110 \end{bmatrix}$$

$$\boxed{\frac{\partial J}{\partial m} = \begin{bmatrix} 45 \\ 113 \end{bmatrix}}$$

* gradient for b

$$\frac{\partial J}{\partial b} = \frac{2}{N}(\hat{y} - y) = \frac{2}{2}\begin{bmatrix} 1 \\ 11 \end{bmatrix} = 1 \times \begin{bmatrix} 1 \\ 11 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$$\boxed{\frac{\partial J}{\partial b} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}}$$

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4. Update parameters m and b

New = old - (α x gradient)

* Update m

$$m = \begin{bmatrix} -1 \\ 2 \end{bmatrix} - 0.01 \begin{bmatrix} 45 \\ 113 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} - \begin{bmatrix} 0.45 \\ 1.13 \end{bmatrix} = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}$$

$$\boxed{m = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}}$$

* Update b

$$b = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 0.01 \begin{bmatrix} 1 \\ 11 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 0.01 \\ 0.11 \end{bmatrix} = \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix}$$

$$\boxed{b = \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix}}$$

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1st iteration

* features $x = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$

* targets $y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$

* weights $m = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}$

* bias $b = \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix}$

① Compute predicted values $\hat{y}$

$$\hat{y} = mx + b$$

* calculate mx

$$mx = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.45 \\ 0.85 \end{bmatrix} = \begin{bmatrix} 1\times(-1.45) + (3\times 0.87) \\ 4(-1.45) + (10\times 0.87) \end{bmatrix} = \begin{bmatrix} -1.45 + 2.61 \\ -5.80 + 8.70 \end{bmatrix}$$

$$\boxed{mx = \begin{bmatrix} 1.16 \\ 2.90 \end{bmatrix}}$$

$$\hat{y} = mx + b = \begin{bmatrix} 1.16 \\ 2.90 \end{bmatrix} + \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix} = \begin{bmatrix} 2.15 \\ 3.79 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} 2.15 \\ 3.79 \end{bmatrix}$$

* Calculate error $\hat{y} - y$

$$\hat{y} - y = \begin{bmatrix} 2.15 \\ 3.79 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix}$$

* MSE: $J = \frac{1}{2}\left((1-1.7851)^2 + (1.3447)^2\right)$

$$J = \frac{13.0066}{2}$$

$$\boxed{J = 6.5033}$$

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② Calculate numerical gradients

* gradient for m

$$\frac{\partial J}{\partial m} = \frac{2}{N} X^T (\hat{y} - y), \quad X^T = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix}$$

Since $N = 2$

$$\frac{\partial J}{\partial m} = \frac{2}{2} \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \times (-2.85) + (4 \times (-2.21)) \\ 3(-2.85) + (10\ (-2.21)) \end{bmatrix}$$

$$= \begin{bmatrix} -2.85 - 8.84 \\ -8.85 - 22.10 \end{bmatrix}$$

$$\boxed{\frac{\partial J}{\partial m} = \begin{bmatrix} -11.69 \\ -30.65 \end{bmatrix}}$$

* gradient for b

$$\frac{\partial J}{\partial b} = \frac{2}{N} (\hat{y} - y)$$

$$\frac{\partial J}{\partial b} = \frac{2}{2} \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix} \Rightarrow \boxed{\frac{\partial J}{\partial b} = \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix}}$$

③ Update parameters m and b

New = old − (α × gradient) with $\alpha = 0.01$

* Update m

$$m = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} - 0.01 \begin{bmatrix} -11.69 \\ -30.65 \end{bmatrix} = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} - \begin{bmatrix} -0.1169 \\ -0.3065 \end{bmatrix}$$

$$= \begin{bmatrix} -1.45 + 0.1169 \\ 0.87 + 0.3065 \end{bmatrix}$$

$$\boxed{m = \begin{bmatrix} -1.3331 \\ 1.1765 \end{bmatrix}}$$

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* Update b.

$$b = \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix} - 0.01 \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix} = \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix} - \begin{bmatrix} -0.0285 \\ -0.0221 \end{bmatrix}$$

$$= \begin{bmatrix} 0.99 + 0.0285 \\ 0.89 + 0.0221 \end{bmatrix}$$

$$\boxed{b = \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix}}$$

End of iteration 2

$$\boxed{m = \begin{bmatrix} -1.3331 \\ 1.1765 \end{bmatrix}}$$

$$\boxed{b = \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix}}$$

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2nd iteration:

* features $X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$

* targets $y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$

* weights $m = \begin{bmatrix} -1.3331 \\ 1.1765 \end{bmatrix}$

* bias $b = \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix}$

1. Compute predicted values

$\hat{y} = mx + b$

* calculate mx

$$mx = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.3331 \\ 1.1765 \end{bmatrix} = \begin{bmatrix} 1\times(-1.3331) + (3\times 1.1765) \\ 4\times(-1.3331) + (10\times 1.1765) \end{bmatrix}$$

$$mx = \begin{bmatrix} -1.333 + 3.5295 \\ -5.3324 + 11.7650 \end{bmatrix}$$

$$mx = \begin{bmatrix} 2.1964 \\ 6.4326 \end{bmatrix} //$$

* $\hat{y} = mx + b = \begin{bmatrix} 2.1964 \\ 6.4326 \end{bmatrix} + \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix}$

$$= \begin{bmatrix} 2.1964 + 1.0185 \\ 6.4326 + 0.9121 \end{bmatrix} \quad \hat{y} = \begin{bmatrix} 3.2149 \\ 7.3447 \end{bmatrix}$$

* Calculate Error

$$\hat{y} - y = \begin{bmatrix} 3.2149 \\ 7.3447 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$\hat{y} - y = \begin{bmatrix} -1.7851 \\ 1.3447 \end{bmatrix} //$$

* MSE : $J = \frac{1}{2}\left((-1.7851)^2 + 1.3447^2\right) = \frac{4.99479}{2} = 2.4974$ //

2. Calculate numerical gradients

* gradient for m

$$\frac{\partial J}{\partial m} = \frac{2}{N} X^T (\hat{y} - y), \quad X^T = \begin{pmatrix} 1 & 4 \\ 3 & 10 \end{pmatrix}$$

$$\frac{\partial J}{\partial m} = \frac{2}{2} \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{pmatrix} -1.7851 \\ 1.3447 \end{pmatrix} = \begin{pmatrix} 1 \times (-1.7851) + (4 \times 1.3447) \\ 3 \times (-1.7851) + (10 \times 1.3447) \end{pmatrix}$$

$$\frac{\partial J}{\partial m} = \begin{pmatrix} -1.7851 + 5.3788 \\ -5.3553 + 13.447 \end{pmatrix} = \begin{bmatrix} 3.5937 \\ 8.0917 \end{bmatrix}$$

gradient for b

$$\frac{\partial J}{\partial b} = \frac{2}{N} (\hat{y} - y) \Rightarrow \frac{2}{2} \begin{pmatrix} -1.7851 \\ 1.3447 \end{pmatrix}$$

$$\frac{\partial J}{\partial m} = \begin{bmatrix} -1.7851 \\ 1.3447 \end{bmatrix}$$

3. update parameters m & b

new = old − (α * gradient) , $\alpha = 0.01$

* update m

$$m = \begin{bmatrix} -1.3331 \\ 1.1765 \end{bmatrix} - 0.01 \begin{bmatrix} 3.5937 \\ 8.0917 \end{bmatrix} = \begin{bmatrix} -1.3331 \\ 1.1765 \end{bmatrix} - \begin{bmatrix} 0.0359 \\ 0.0809 \end{bmatrix}$$

$$m = \begin{bmatrix} -1.3690 \\ 1.0956 \end{bmatrix} //$$

* update b

$$b = \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix} - 0.01 \begin{bmatrix} -1.785 \\ 1.3447 \end{bmatrix} = \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix} - \begin{bmatrix} -0.0179 \\ 0.0134 \end{bmatrix}$$

$$= \begin{bmatrix} 1.0185 + 0.0179 \\ 0.9121 - 0.0134 \end{bmatrix} \Rightarrow b = \begin{bmatrix} 1.0364 \\ 0.8987 \end{bmatrix} //$$

Trend of m, b and the Error

(1) m and b

The weights do not move in one direction. Update 1 is a large jump (m_2 falls from 2 to 0.87) because the starting gradient was large ([45, 113]). The step overshoots the minimum, so the next gradient has the opposite sign and m_2 bounces back to 1.18, then settles at 1.10. m_1 does the same: $-1 \to -1.45 \to -1.33 \to -1.37$. The steps shrink each time (m_2 moves ~~from~~ 1.13, 0.31, then 0.8), so the oscillation is damping out.

The bias changes far less. It is because ~~started~~ $\frac{\partial J}{\partial b} = \frac{2}{N}(\hat{y} - y)$ uses the error alone, while $\frac{\partial J}{\partial m} = \frac{2}{N} X^T(\hat{y} - y)$ multiples the error by the features. That make m's gradient much larger, so m moves more under the same learning rate.

(2) Is the error going down?

Yes. J fell from 61 → 6.50 → 2.50 → 2.16 over the updates, so each update moved the parameters in a direction that reduced the cost. The drops get smaller (54.5, 4.0, 0.33), which means the steps are becoming more careful as we approach a minimum. Three updates are not enough to converge: The final predictions are [2.85, 6.38) against targets [5, 6], so more iterations are needed.

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